APEX VELO & CO: PERFORMANCE INTELLIGENCE REPORT

Abstract

This report analyzes the commercial performance of **Apex Velo & Co.** using transactional sales data. The analysis evaluates revenue trends, product performance, regional contributions, and pricing dynamics to assess overall business health. It highlights key revenue drivers, seasonal demand patterns, and product mix effects that influence profitability and growth. The goal is to translate these findings into practical insights that support pricing strategy, product positioning, and operational planning.

APEX VELO& CO

Introduction

Apex Velo & Co. is a premier global manufacturer and multi-channel retailer of high-performance cycling equipment and outdoor gear. Headquartered in a major metropolitan hub, the company operates across 10 international territories, catering to a diverse demographic ranging from professional athletes to weekend enthusiasts.

Insights and recommendations are provided on following key areas :

Revenue Performance Analysis
Evaluates revenue patterns across customers, products, and territories to identify the primary sources of sales growth.

Product Performance Insights
Examines category and product-level performance to determine which products contribute most to revenue.

Regional Market Analysis
Compares territory performance to identify the strongest and weakest markets.

Operational Efficiency Assessment
Evaluates sales and store performance to understand operational contribution to revenue.

Profitability and Margin Analysis
Investigates pricing and margin patterns to detect potential profitability risks.

Links to work given below:

Access the whole Project Here

An interactive PowerBI dashboard can be downloaded Here

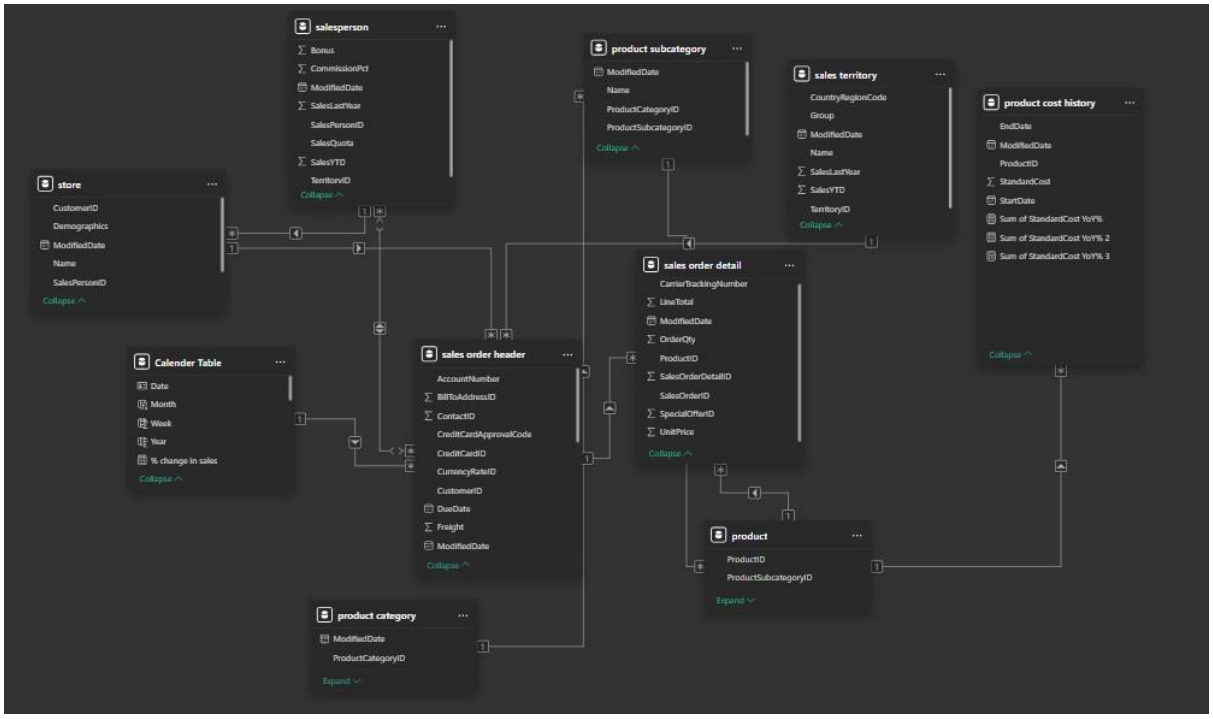
All SQL queries can be found Here

SQL queries regarding Data Validation can be found Here

SQL queries regarding Business Questions can be found Here

Readme page can be accessed Here

Given below are the resources used for making the analysis Data Structure :



Details

The dataset follows a **relational sales model** centered around transactional order data. Sales tables capture order-level details such as product, quantity, price, and order dates, forming the core of the analysis.

Supporting tables provide additional business context, including **product information (categories and subcategories), customer details, store**

and salesperson data, and geographic territories. These relationships allow sales performance to be analyzed across **products, customers, locations, and time**, enabling comprehensive revenue and operational insights.

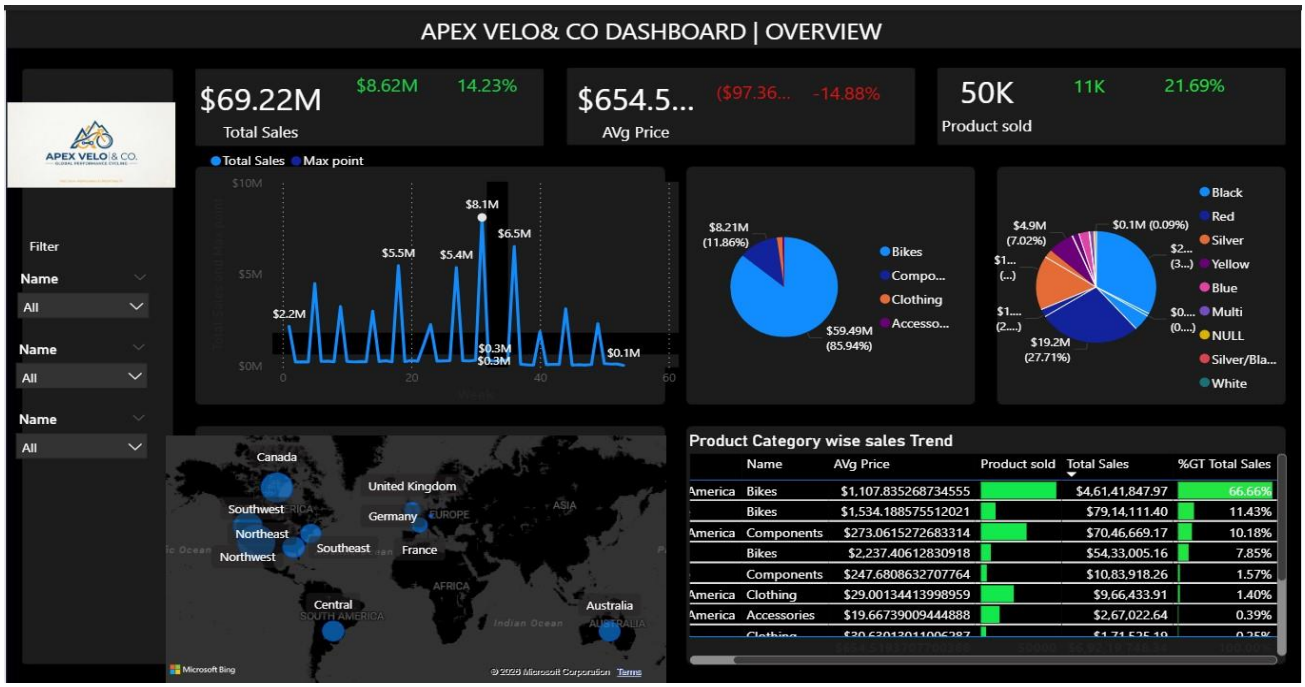
Executive Summary

Overview of Findings:

Year-over-year sales increased by **$8.62M**, representing **14.23% revenue growth**. However, this growth coincides with a **14.88% decline in average product prices**. At the same time, **product sales volume increased by 21.69%**.

This suggests that recent growth is primarily driven by **higher sales volume rather than stronger pricing power**.

Category analysis shows that **Bikes account for the majority of revenue**, indicating a strong dependence on a single product category.



Further analysis is as follows:

Sale Trends

While **Bikes** dominate the total sales value (85.94% as seen in the blue pie chart), the **Product Category wise sales Trend** table reveals a massive disparity in volume versus value:

Revenue Concentration

Bikes contribute **85.94% of total revenue**, making them the primary driver of company sales. North America is the dominant market, accounting for **66.66% of total sales**.

This concentration introduces risk because small fluctuations in bike sales could significantly affect overall revenue.

Margin vs Volume Trade-Off

Although total product sales increased significantly, the **average selling price declined by $97.37**. This pattern suggests that revenue growth may be driven by **discounting strategies or a shift toward lower-priced products**.

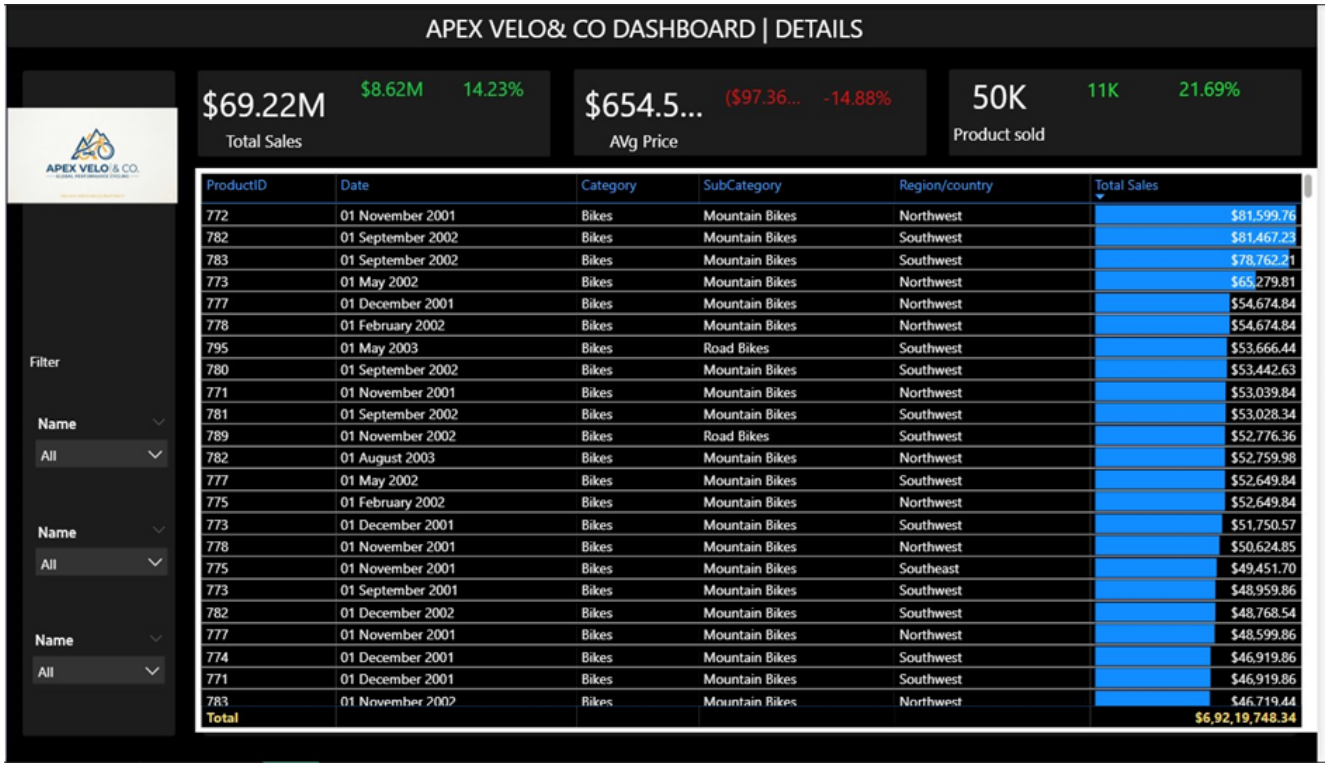
Seasonal Volatility

Weekly revenue patterns show **sharp peaks and troughs**, including major spikes such as **$8.1M in week 38**. These fluctuations indicate that sales are heavily influenced by **specific seasonal events or product launches** rather than stable baseline demand.

Customer Preferences

Product color analysis shows that **Black and Red products represent nearly 40% of total sales**. These colors appear to be the most reliable inventory choices, while colors such as **White and Multi-color represent smaller niche segments**.

A Deeper Dive into the Data



ProductID	Date	Category	SubCategory	Region/country	Total Sales
772	01 November 2001	Bikes	Mountain Bikes	Northwest	$81,599.76
782	01 September 2002	Bikes	Mountain Bikes	Southwest	$81,467.23
783	01 September 2002	Bikes	Mountain Bikes	Southwest	$78,762.21
773	01 May 2002	Bikes	Mountain Bikes	Northwest	$65,279.81
777	01 December 2001	Bikes	Mountain Bikes	Northwest	$54,674.84
778	01 February 2002	Bikes	Mountain Bikes	Northwest	$54,674.84
795	01 May 2003	Bikes	Road Bikes	Southwest	$53,666.44
780	01 September 2002	Bikes	Mountain Bikes	Southwest	$53,442.63
771	01 November 2001	Bikes	Mountain Bikes	Northwest	$53,039.84
781	01 September 2002	Bikes	Mountain Bikes	Southwest	$53,028.34
789	01 November 2002	Bikes	Road Bikes	Southwest	$52,776.36
782	01 August 2003	Bikes	Mountain Bikes	Northwest	$52,759.98
777	01 May 2002	Bikes	Mountain Bikes	Southwest	$52,649.84
775	01 February 2002	Bikes	Mountain Bikes	Northwest	$52,649.84
773	01 December 2001	Bikes	Mountain Bikes	Southwest	$51,750.57
778	01 November 2001	Bikes	Mountain Bikes	Northwest	$50,624.85
775	01 November 2001	Bikes	Mountain Bikes	Southeast	$49,451.70
773	01 September 2001	Bikes	Mountain Bikes	Southwest	$48,959.86
782	01 December 2002	Bikes	Mountain Bikes	Southwest	$48,768.54
777	01 November 2001	Bikes	Mountain Bikes	Northwest	$48,599.86
774	01 December 2001	Bikes	Mountain Bikes	Southwest	$46,919.86
771	01 December 2001	Bikes	Mountain Bikes	Southwest	$46,919.86
783	01 November 2002	Bikes	Mountain Bikes	Northwest	$46,719.44
Total					$6,92,19,748.34

Product Performance

The data indicates strong specialization within the Bikes category. **Mountain Bikes dominate high-value transactions**, representing the majority of the top revenue-generating sales.

This confirms that Mountain Bikes are a key component of the company's premium product strategy.

Regional Sales Concentration

High-value transactions occur most frequently in the **Northwest and Southwest regions**, suggesting that these markets are particularly receptive to premium bike models.

Price Point Distribution

Individual high-value transactions range between **$40K and $81K**, indicating that a small number of premium bike purchases contribute significantly to total revenue.

Although the overall average product price is **$654.52**, these premium transactions form the foundation of the company's **$69.22M total revenue**.

Historical Data Context

The dataset spans transactions between **2001 and 2003**, capturing a historical period of growth. Comparing the frequency of high-value transactions across years provides insight into changes in premium product demand.

Discount analyses

Business Insights

- **Low Price Sensitivity**: Customers aren't buying significantly more just because of a discount. A 5% discount moves almost the same amount of product as a 2% discount.
- **The 40% "Sweet Spot":** Only deep discounts (40%) show a clear jump in order volume. Anything less than that is likely giving away profit to people who would have bought the items anyway.
- **Concentration Risk**: The business relies almost entirely on "Bikes" in North America. This is a "Danger Zone"—if bike demand drops, the company has no other strong revenue source to fall back on.

Recommendations:

1. Margin Protection & Pricing Strategy

- **Audit Discounting Practices:** With **Average Price** down by **14.88%** despite a **21.69%** increase in volume, investigate if aggressive discounting is eroding profitability.
- **Premium Upselling:** Since **Mountain Bikes** represent your highest-value transactions (reaching up to **$81.5k** per sale), create bundled "Premium Maintenance" packages to increase the lifetime value of these high-ticket customers.

2. Revenue Diversification

- **Improve Attachment Rates: Accessories** and **Clothing** account for high volume but less than **2%** of total revenue. Implement "Frequently Bought Together" prompts for customers purchasing **Bikes** to increase the share of these categories.
- **Expand Road Bike Presence:** The Details page shows **Road Bikes** (e.g., Product 795) generating significant single-sale revenue. Expanding this sub-category could reduce the heavy reliance on **Mountain Bikes**.

3. Inventory & Operational Efficiency

- **Stock Optimization by Colour:** Prioritize inventory levels for **Black and Red** variants, as they constitute nearly **40%** of total sales.
- **Seasonal Leveling:** The "Total Sales" line chart shows extreme volatility with sharp peaks (e.g., **$8.1M**) and deep valleys. Launch mid-season loyalty campaigns or "trade-in" events to stabilize revenue during the "lull" weeks.

4 . Stop Small Discounts: Eliminate all discounts below 10%. They don't drive enough new sales to justify the loss in profit.

- **Use Bundles, Not Price Cuts:** Instead of lowering the price of a bike, offer a "Starter Kit" bundle (Bike + Helmet + Lights). This keeps your bike margins high while moving lower-performing accessories.
- **Focus on Diversification:** Use promotions specifically to grow categories like "Clothing" or "Accessories" in territories outside of North America to reduce your risk**.**

5. A Potential Danger Zone to be Looked For

One of the most notable findings is the high concentration of revenue within a single product segment.

The Bikes category contributes approximately 85.94% of total revenue, with North America alone accounting for nearly 66.66% of sales within this segment. While this indicates strong demand for premium bicycles, it also exposes the company to significant concentration risk.

Such dependence on a single product line creates financial vulnerability. Any disruption affecting this segment—such as market saturation, supply chain constraints, economic downturns, or competitive pressure—could have a disproportionate impact on overall company revenue.

From a financial resilience perspective, this level of concentration suggests that the company's **revenue stability is strongly tied to the continued success of one core product segment**. Without greater diversification across product lines, the business may face **elevated risk exposure if demand for bikes weakens in key markets**.